BIOS V2 · Thapar Institute · Prune track: Ship It Lean

SafeTrail

Warn early. Never lose an SOS. Prove every response.

A landslide on the trail. No signal, no common language, and a crowd building behind you. In the mountains, everything fails at once.

Where it breaks

No warning
Walking into closed or unsafe stretches.

No signal
Calls fail exactly where help is needed.

No common language
A pilgrim can't say where, or what.

Why it matters

16.5 lakh+
pilgrims visited Kedarnath in 2025, counted by early October
ETV Bharat, Oct 2025

15,000+
evacuated from the Kedarnath trek after the July 2024 cloudburst
Deccan Herald, Aug 2024

Up to 3,000
people stranded at Lachen and Lachung, North Sikkim, Oct 2023
Telangana Today (ANI), Oct 2023

Pilots: Kedarnath · Lachung to Yumthang, Sikkim · Corbett Tiger Reserve



A landslide blocks the only road. Far-western Nepal.
Photo: Possible (Flickr), CC BY 2.0

For the tourist: warned early, never alone

Before you go

- Join by QR code, no app store
- Safety pack in your language
- Signed trip pass that works offline

On the trail

- Danger-zone alerts with no internet
- Crowd warnings before you walk in
- Weather by section, safe way back

In trouble

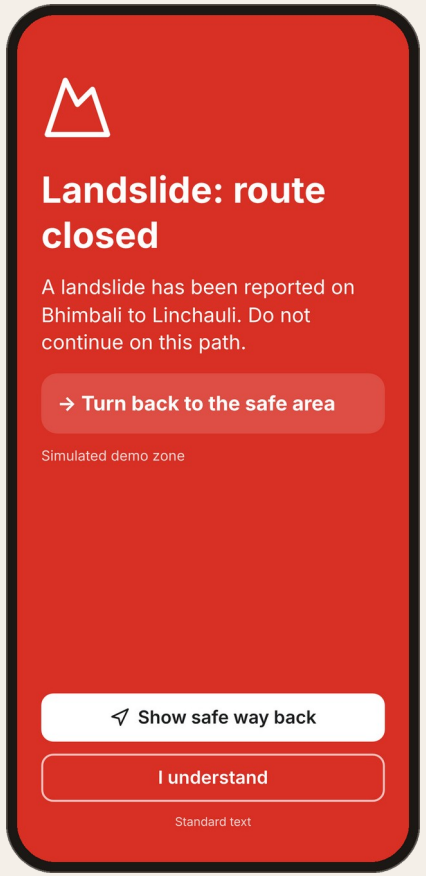
- Hold SOS: three taps, no typing
- Saved on the phone, sent on first signal
- Whistle, flash and a read-to-112 card

Help is coming

- “Help acknowledged” on every phone
- “Are you safe?” after a disaster
- Family link: status, never location

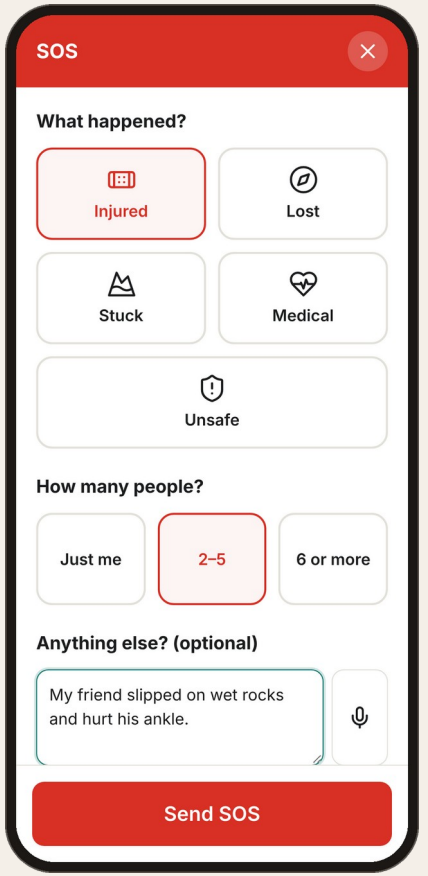
Photo: Steve Hicks, trekking in the upper Khumbu, Nepal (CC BY 2.0)

Real screens from the app: one installable web app that works offline once opened.



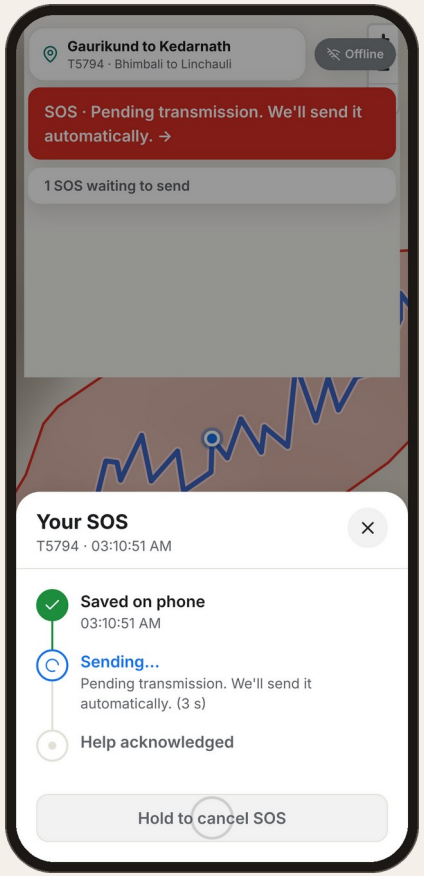
Danger-zone alert

Full screen, from the phone's own check. No signal needed.



SOS in three taps

What happened and how many. No typing, in any language.



Offline, still safe

Saved with time and place. Sends itself on the first signal.

For responders: 40 calls become one clear incident

The same app with an officer login. Counts come from code; the AI only writes.

SOS arrives

By data, coded SMS or at a checkpoint. Duplicates dropped.

Grouped by place

SOS within 150 m and 30 min: one incident, exact counts.

AI report via Prune

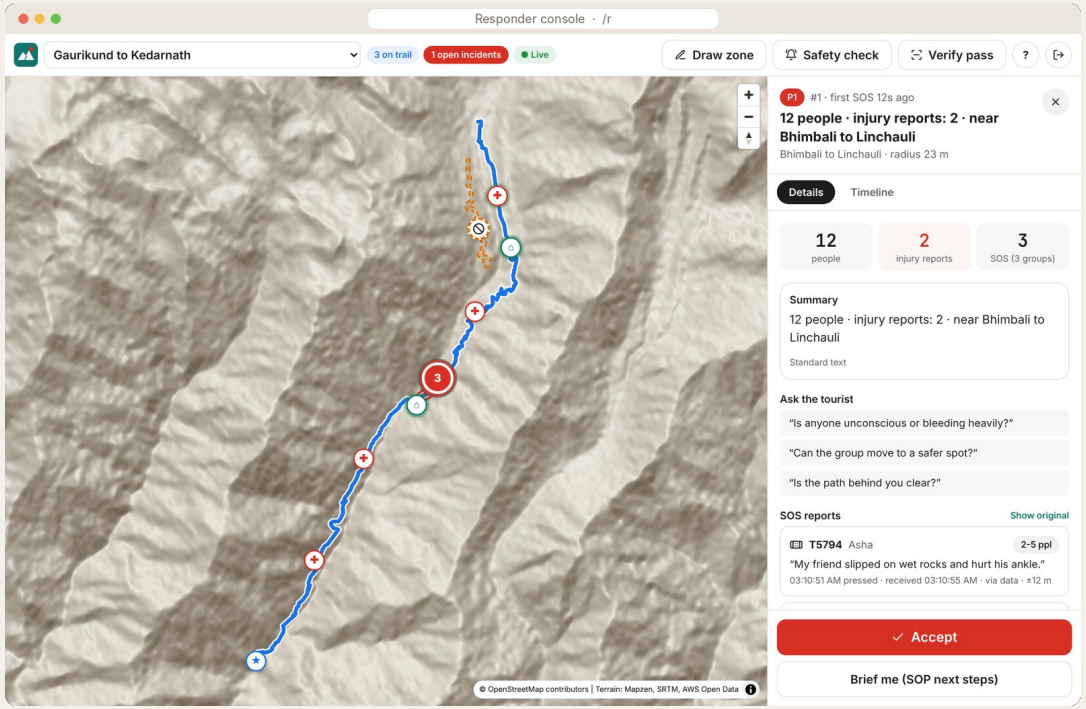
Summary and notes in the officer's language.

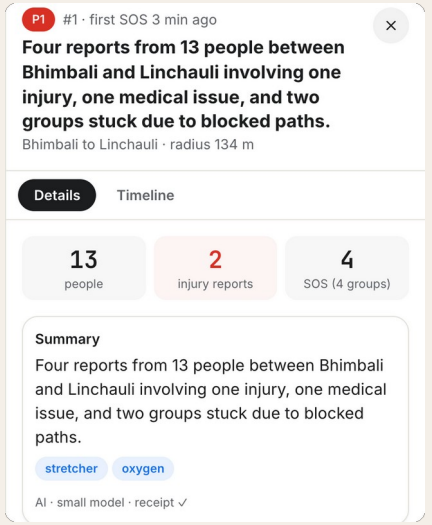
Accept

Every phone in the group shows “Help acknowledged”.

Signed timeline

Every step hash-chained and signed. Edits show.





Live AI summary via Prune, with its signed receipt (Hardik's laptop).



One clear picture for the rescue team, not 40 calls.

Photo: ResoluteSupportMedia, mountain rescue training (CC BY 2.0)

Crowd heat + weather

“Are you safe?” tally

Draw a zone

Verify passes

The AI safety pack: made once per route, read offline

Written through Prune before the trip, checked by validators, shared by every tourist on that route and language.

What's inside

Route briefing
Distance, altitude, the emergency number and control room.

Risks by section
What each stretch is like, and the nearest help point.

Altitude sickness
The warning signs, and what to do right now.

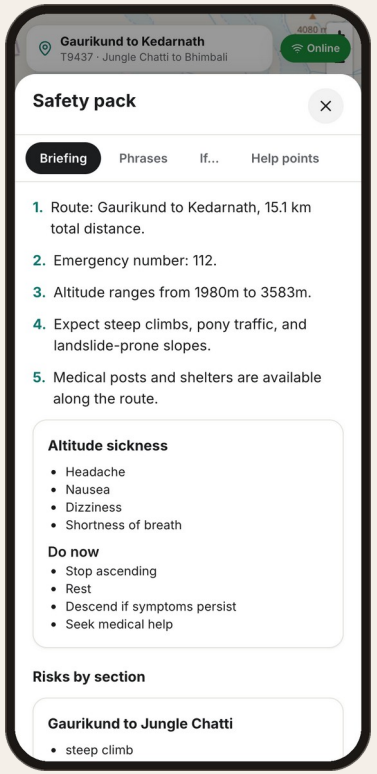
Phrases and "Show to a local"
Hindi or Nepali, read aloud or shown full screen.

Help points
A compass guide to the nearest shelter, offline.

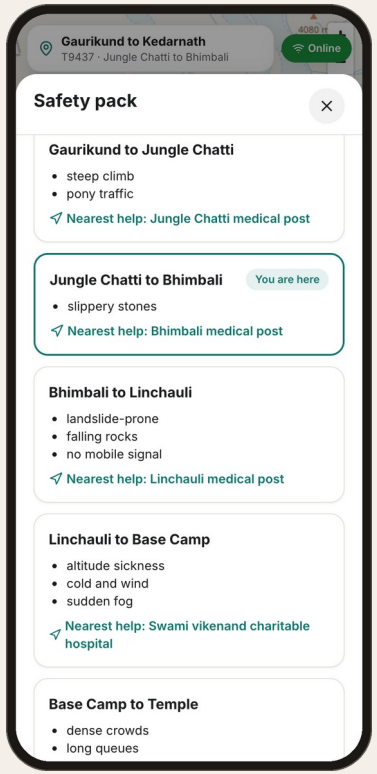
How it's made

Verified route facts and SOPs → **Prune: small model first** → Validators: no invented facts → Cached per route and language

Pack #2 onward costs nothing: it is shared, not regenerated. If the AI is down, the phone shows "Standard text" written by us.



Route briefing
Real AI output through Prune (Hardik's laptop).



Risks by section
Marks the stretch you are on.



For every pilgrim
Photo: McKay Savage, Haridwar (CC BY 2.0)

Also written by AI through Prune: SOS notes translated · Group summaries · SOP next steps · Zone warnings per language · Free-text SMS · Permit explainer

Your pass comes from the permit you already have

Phase 2, built with a simulated registry: link the e-Visa, e-ILP or yatra registration you already hold.

Foreign visitor — e-Visa or visa number, plus PAP or RAP where the state requires one

Indian visitor — e-ILP, yatra registration, DigiLocker, passport or voter ID

Aadhaar — Optional: the last 4 digits only, never required

How linking works

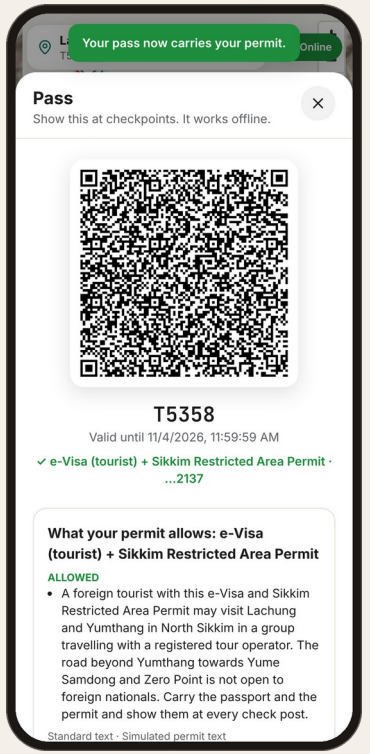
1 Enter the visa or permit number → 2 Checked against the registry → 3 Hashed at once; only the last 4 kept → 4 Pass re-signed with permit and allowed zones

Checked offline at any checkpoint

Valid · Expired · Forged · Revoked · Not permitted here

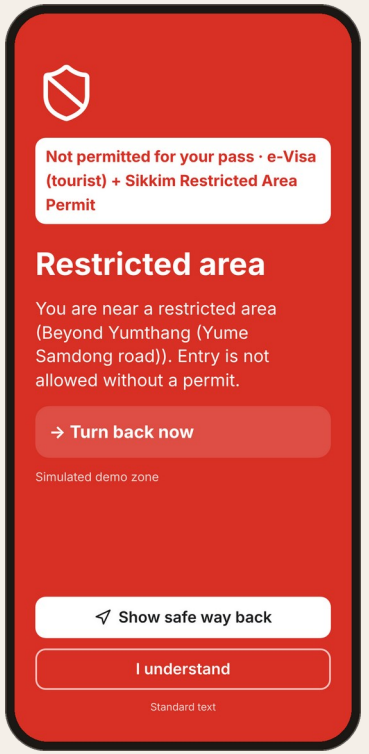
The full number is never stored, logged or sent to the AI: only a salted hash and the last 4 characters. Upgrades and revocations are signed into the event log, and checkpoints keep the revocation list offline.

Hackathon build: a simulated registry with fake numbers. A real rollout plugs into the official e-Visa, e-ILP and yatra checks.



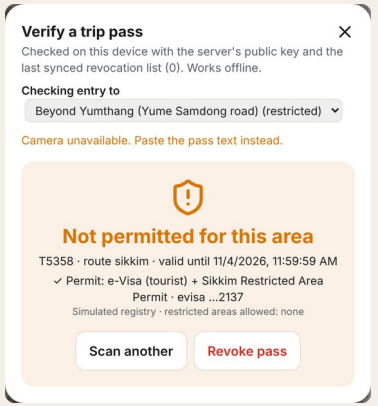
Permit linked

e-Visa + Sikkim permit, last 4 only.



Past the permit's limit

Walking toward the road beyond Yumthang.



Checkpoint scan, offline: "Not permitted for this area".



North Sikkim: where you may go depends on your permit.
Photo: genobz, Sikkim (CC BY 2.0)

How it's built, and what it costs

Safety runs as plain code on the phone and the server. Every AI call goes through Prune.

Phone, works offline
Zone alerts, SOS queue, pass check, phrases

→

Server: FastAPI + SQLite
Grouping, counts, signed log, live updates

→

Prune gateway
Budgets, cache, injection check, receipts

→

Gemini models
3.1 Flash-Lite (small), 3.5 Flash (large)

The AI is never in the emergency path: if it's down, every alert and SOS still works.

React + TypeScript PWA (service worker, IndexedDB queues) · MapLibre + OpenStreetMap · FastAPI · SQLite · Server-Sent Events · Ed25519-signed passes · hash-chained event log · Open-Meteo weather

What Prune does for us

- One base URL. A checker fails the build on any direct AI call.
- Small model by default, the large one only when needed.
- Budgets: ours, then Prune's run budget and step limit, then a daily key cap.
- Exact cache, made safe with a request fingerprint after we caught it mixing up incidents.
- Injection checks on SOS notes. No personal data in prompts.
- A signed receipt on every call.

Measured on 100 simulated trips

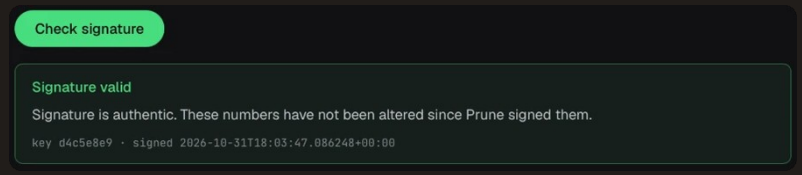
82% fewer
tokens per trip than the naive way

≈ 3 paise
per monitored trip, at Google's list price

100%
of counts correct: code, never AI

72%
of repeat-day calls served from Prune's cache

Same 100-trip scenario run four ways; costs from Google's own token counts at list price (docs/BENCHMARK.md).



withprune.com/verify · 77 backend + 6 app tests · 10 of 11 Prune checks



SafeTrail doesn't replace 112. It's the safety layer before it.

Who pays: state tourism departments, pilgrimage boards and park authorities, with a licence and a small fee per tourist.

github.com/gaurvit06/Safetrail

Photo: Matt Zimmerman (CC BY 2.0)